THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 7: **Particulate Nature of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Hot tea cools faster when poured in a wide saucer. Reason (R): A larger surface area increases the rate of evaporation. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. 100 mL of water and 100 mL of ethanol are mixed. The volume of the mixture is slightly less than 200 mL. This best illustrates: 1

(a) sublimation **(b)** evaporation
(c) condensation **(d)** spaces between particles

3. A student observes that a drop of food colour spreads equally in a glass of still water but faster in warm water. The explanation is: 1

(a) colour is lighter
(b) temperature increases the motion of particles
(c) gravity acts differently
(d) water is thicker

4. Assertion (A): A gas can be compressed easily but a solid cannot. Reason (R): Gas particles are widely separated while solid particles are tightly packed. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. A sample of a pure solid is heated at a constant rate and the temperature is recorded. A horizontal portion on the graph shows: 1

(a) freezing only **(b)** change of state **(c)** chemical change **(d)** boiling only

6. In a closed bottle half-filled with water at constant temperature, the rate of evaporation compared with the rate of condensation is: 1

(a) greater always **(b)** less always
(c) zero **(d)** equal at equilibrium

7. Assertion (A): The temperature stays constant while ice melts. Reason (R): The heat supplied is used to increase the kinetic energy of particles. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. Ice at 0 degrees Celsius and water at 0 degrees Celsius are compared. Which statement is correct? 1

(a) Ice particles move faster
(b) Water has fewer particles
(c) Water has lower temperature
(d) Particles have the same kinetic energy but different arrangement

9. Which has the weakest forces between its particles at room temperature? 1

(a) ice **(b)** water **(c)** iron **(d)** oxygen

10. Assertion (A): A steam burn is more severe than a burn from boiling water. Reason (R): Steam gives out extra heat on condensing. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain why a gas does not collect at the bottom of a container although it has weight. 2

12. Why does sweating not cool us effectively when the humidity is high? 2

13. State why evaporation is called a surface phenomenon. 2

14. A beaker of crushed ice at 0 degrees Celsius is heated steadily. What happens to the thermometer reading while the ice melts, and where does the heat go? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A student heats 50 g of ice from -10 degrees Celsius. Describe, with temperature values, the four stages of the heating curve till 110 degrees Celsius. 3

16. Compare ice, water and steam in a table in terms of energy, spacing and motion of particles. 3

17. How would the boiling point change if you added salt to water? Explain briefly. 3

OR

Explain the statement: 'Temperature is a measure of the average kinetic energy of particles' using hot and cold water. 3

18. Why does the mass of a solid not change on melting although its volume may change? 3

OR

A boy says that 'cold things have no heat'. Correct this using the particle model. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Design an investigation to find how the temperature affects the rate of diffusion of ink in water. Mention variables, apparatus, observations and conclusion. 5

OR

Calculate: how many times does the volume increase when 1 mL of water changes to steam if the volume of steam is about 1600 mL at 100 degrees Celsius? Explain the reason. 5

20. Analyse the cooling curve of molten wax, naming each stage and explaining the flat part in terms of particles and energy. 5

OR

Explain how the combination of pressure and temperature determines whether a substance is solid, liquid or gas with examples of CO_2 and water. 5

END OF QUESTION PAPER — Total: 40 Marks